

Limited spatial transfer of a reusable CA1 wall-related population update

An exploratory longitudinal reanalysis

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Abstract

Environmental boundaries strongly influence hippocampal place fields, but it is unresolved whether a changed boundary evokes only a location-specific map change or also a neural population update that can be reused elsewhere. We reanalysed longitudinal calcium-imaging data from 5,413 dorsal CA1 neurons in seven mice across repeated exposure cycles containing a familiar square and nine internal-wall geometries. For each registered cell, we learned a local wall-minus-open response contrast from training geometries and asked whether that contrast predicted a later, held-out response. At the same physical seam, the learned contrast predicted held-out wall responses better than open responses across all animals (252 predictions; mean correlation advantage 0.184). A single-tile counterfactual, in which two environments differed only at one lattice location, produced the same result (46 predictions; mean advantage 0.225; seven of seven animals), including after adjustment for speed, heading, time, and sampling. More importantly, a contrast learned at a non-overlapping source seam predicted a later wall response at a target seam 25 cm away ($\bar{\rho} = 0.148$; seven of seven animals). This shifted prediction was weaker than a query-matched exact-location benchmark ($\bar{\rho} = 0.230$; transfer-minus-exact = -0.082) and remained positive after behavioral adjustment ($\bar{\rho} = 0.119$). A stringent same-session mirror-open control favored the wall target on its supported, demeaned tangential subset (mean advantage 0.114; five of five eligible animals), but support exclusions and weaker raw correlations limit its scope. Cell-identity permutations, direction controls, and the small number of factorial wall combinations further constrained interpretation. We conclude that CA1 wall-related remapping contains a cell-resolved component that is reusable across global layouts and partially transferable across nearby locations, but is not a location-invariant or demonstrably compositional primitive. The result motivates a prospective experiment that independently randomizes wall location, orientation, order, and experience.

Keywords: hippocampus; CA1; place cells; boundaries; population code; transfer; remapping; cognitive map

Significance statement

Mental maps are often described as reusable, but that phrase can mean anything from revisiting a familiar place to applying the same neural change at a new location. We test the stronger meaning directly. A cell-by-cell CA1 response learned when a wall occupied one location carried information about a later wall response at a different, non-overlapping location. The transfer was reliable across animals but smaller than reuse at the original location, and available controls do not establish a fully

portable or compositional “wall code.” The contribution is therefore a quantitative boundary on reusability: hippocampal map updates can generalize beyond an exact coordinate, while retaining substantial location dependence.

1 Introduction

Place cells relate an animal’s position to a distributed hippocampal activity pattern, yet the map is not a passive image of Euclidean space. Place fields are anchored by environmental geometry, shift when boundaries move, and can appear at characteristic distances and directions from walls (O’Keefe and Burgess, 1996; Hartley et al., 2000; Barry et al., 2006). Boundary-responsive inputs provide one mechanistic route by which local geometry could organize hippocampal representations (Lever et al., 2009). More generally, current accounts treat hippocampal-entorhinal codes as structured representations whose parts may be recombined or generalized across experiences (Stachenfeld et al., 2017; Wernle et al., 2018; Bakermans et al., 2025).

These observations leave an important distinction unresolved. A wall-related response may recur at the same coordinate despite changes elsewhere in the environment. That is *exact-location reuse*: a local feature of an otherwise changing map remains stable. A stronger possibility is *cross-location transfer*: the cell-resolved response contrast evoked by a wall at one seam predicts the response to a wall at a different seam. This second claim is not implied by ordinary place-field stability or by the existence of boundary-vector-like firing. It requires an out-of-sample test of whether the identities and signed response changes of individual cells carry across spatial coordinates.

The distinction matters for theories of cognitive maps. A perfectly portable boundary operation would resemble a location-invariant representational primitive. A nonzero but attenuated transfer would instead suggest a reusable update embedded within a place-dependent code. Failure to transfer would be consistent with each wall response being reconstructed independently from position, behavior, or the full layout. These alternatives can agree on mean rate maps while differing in the cross-registered population structure.

The dataset of Lee et al. (2025) is unusually useful for separating these possibilities. Seven mice were imaged longitudinally in dorsal CA1 while foraging in a familiar square and in nine environments made by placing internal walls on a common spatial lattice. Repeated exposure cycles provided registered cells, recurring local seams, and later held-out sessions. The design was not created for a transfer test, so the present work is exploratory and constrained by fixed environment order. Its geometry nevertheless permits three progressively stronger questions: does a learned wall-open contrast recur at the same seam across global contexts; does an especially clean single-tile contrast recur; and does a contrast learned at one seam predict a wall response at a neighboring, non-overlapping seam?

We find affirmative answers to all three, with an important gradient. Exact-location reuse is strongest, cross-location transfer is smaller, and available composition and direction controls do not support a fully portable wall operator. The result is therefore neither a restatement that boundaries shape place fields nor evidence for an innate mental-map module. It is a specific population-level observation: part of a local CA1 wall update survives a change of spatial coordinate.

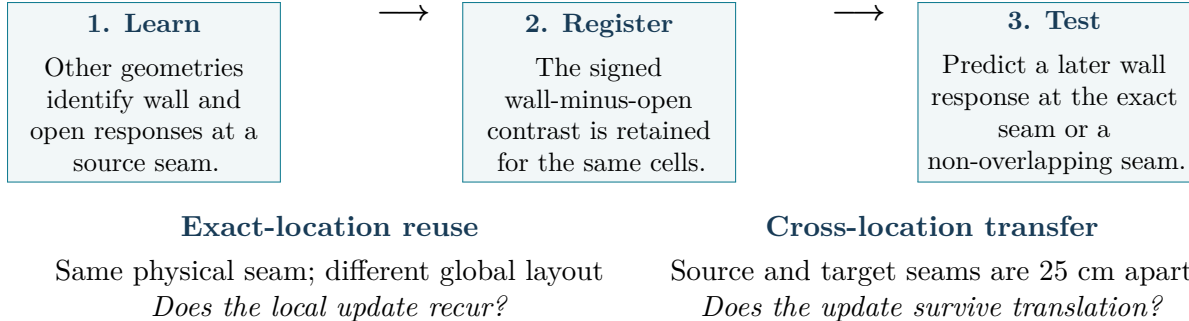


Figure 1: Analysis logic. All tests use registered-cell response vectors learned from training geometries and evaluated on later held-out neural outcomes. Exact reuse and shifted transfer are separate estimators: the cross-location result is not a transported version of the single-tile counterfactual.

2 Results

2.1 A held-out test of cell-resolved wall updates

We represented each candidate wall location as an oriented seam between two adjacent lattice tiles. For a registered cell, an environment, and a seam, we estimated activity in a narrow local strip and subtracted the corresponding estimate from a bracketing familiar-square session. This square residual reduced stable place preference while retaining the environment-specific local change.

Training geometries supplied two cell-wise profiles at each seam: a mean residual when the seam was blocked by a wall and a mean residual when it was open. Their difference was the learned wall-related update. The neural outcome from the later target geometry was never used to form that source contrast. Prediction was scored by a rank correlation across held-out registered cells. We report animal-level means and sign counts rather than treating cells or queries as independent biological replicates.

2.2 A local wall-open contrast recurs across global geometries

We first tested exact-location reuse while matching the broader wall context between training and held-out comparisons. Across 252 held-out predictions, the wall-derived profile correlated more strongly with later wall responses than did the open-derived profile. The mean wall-minus-open correlation advantage was 0.184 and was positive in all seven animals (Fig. 2A). The sign was also positive in all animals when horizontal and vertical seams were evaluated separately.

This analysis establishes more than a population-average increase near walls. The predictive object is a vector indexed by registered cell identity. Cells that contributed positively or negatively to the learned local update tended to do so again when the same seam was encountered in another global geometry. However, the seam itself was unchanged, so this result alone does not show that the update can move through space.

2.3 A one-tile counterfactual sharpens exact-location reuse

Two environments in the sequence, denoted u and o , differed only in whether one lattice tile was blocked. Their contrast therefore isolates one local wall state more cleanly than comparisons among complex layouts. We learned the $u - o$ cell-wise contrast at that focal seam and tested it in held-out i , l , and t geometries that independently instantiated wall or open states there.

Across 46 held-out predictions, the learned vector correlated with wall outcomes ($\bar{\rho} = 0.197$) but not open outcomes ($\bar{\rho} = -0.028$), giving an advantage of 0.225, positive in all seven animals (Fig. 2A). The raw-rate version gave a similar advantage (0.243). A model that adjusted local bin estimates for speed, heading, elapsed time, and occupancy retained the result, both with time included (0.225) and omitted (0.229).

The effect was more evident in the nearest independently eligible distance band, 2.5–7.5 cm from the seam (mean advantage 0.233, seven of seven animals), than in a 17.5–22.5 cm band (0.069, six of seven; Fig. 2D). These bands had separately defined supports and are therefore a descriptive localization comparison, not a paired inferential contrast.

The fixed sequence imposes an important caveat: u always followed o . A generic same-seam, unchanged-state later-minus-earlier placebo was opposite in sign (mean -0.042 , zero of seven positive), whereas the actual $u - o$ contrast was positive (mean 0.195, seven of seven) and remained so after the nuisance adjustment (mean 0.212). This argues against a generic drift explanation, but it cannot remove an environment-specific order effect because wall state and order were never randomized.

2.4 A learned wall update partially transfers to another location

We next asked the stronger question. For each eligible target, we learned a wall-minus-open vector at a distinct source seam with the same signed wall normal. The source and target seams were 25 cm apart, their local analysis bins did not overlap, and source training sessions were disjoint from the target session. The nearest source-target bins were 5 cm apart for tangential translations and 15 cm apart for normal-axis translations. The test used a separate multi-geometry estimator rather than the $u - o$ contrast.

The translated source vector predicted the later target-wall residual with mean correlation 0.148, positive in all seven animals (Fig. 2B). It also outperformed a source-open predictor by 0.152, and was more aligned with the target-wall response than with a training-derived target-open profile by 0.216, again positive in every animal. Thus the result cannot be summarized merely as two wall locations having high activity: a signed, cell-resolved update learned at one coordinate carried information about another.

Transfer was incomplete. On exactly the same eligible queries, cells, and support, an exact-location effect predicted the target response with mean correlation 0.230. The shifted-minus-exact difference was -0.082 , and only one of seven animals favored the shifted predictor. The data therefore support a reusable but spatially biased population update, not a location-invariant wall template.

2.5 Behavior and mirror-open controls constrain smooth-field explanations

Because animals may run differently near internal barriers, we re-estimated local neural response maps with covariates for speed, first and second circular harmonics of heading, elapsed time, and spatial sampling. The behavior-adjusted translated effect remained positive in all seven animals ($\bar{\rho} = 0.119$). Its specificity against the training-derived target-open profile was 0.172, and its advantage over a source-open predictor was 0.138, both seven of seven. Omitting elapsed time changed these values by less than 0.003.

The registered-wall and target-wall strips could nevertheless share generic nearby place-field structure. We therefore defined a stricter within-session control before inspecting its neural endpoint in the project workflow. For each wall target, a mirror-open seam in the same target session was matched for translation distance, axis, signed normal, relative bins, and registered cells. The

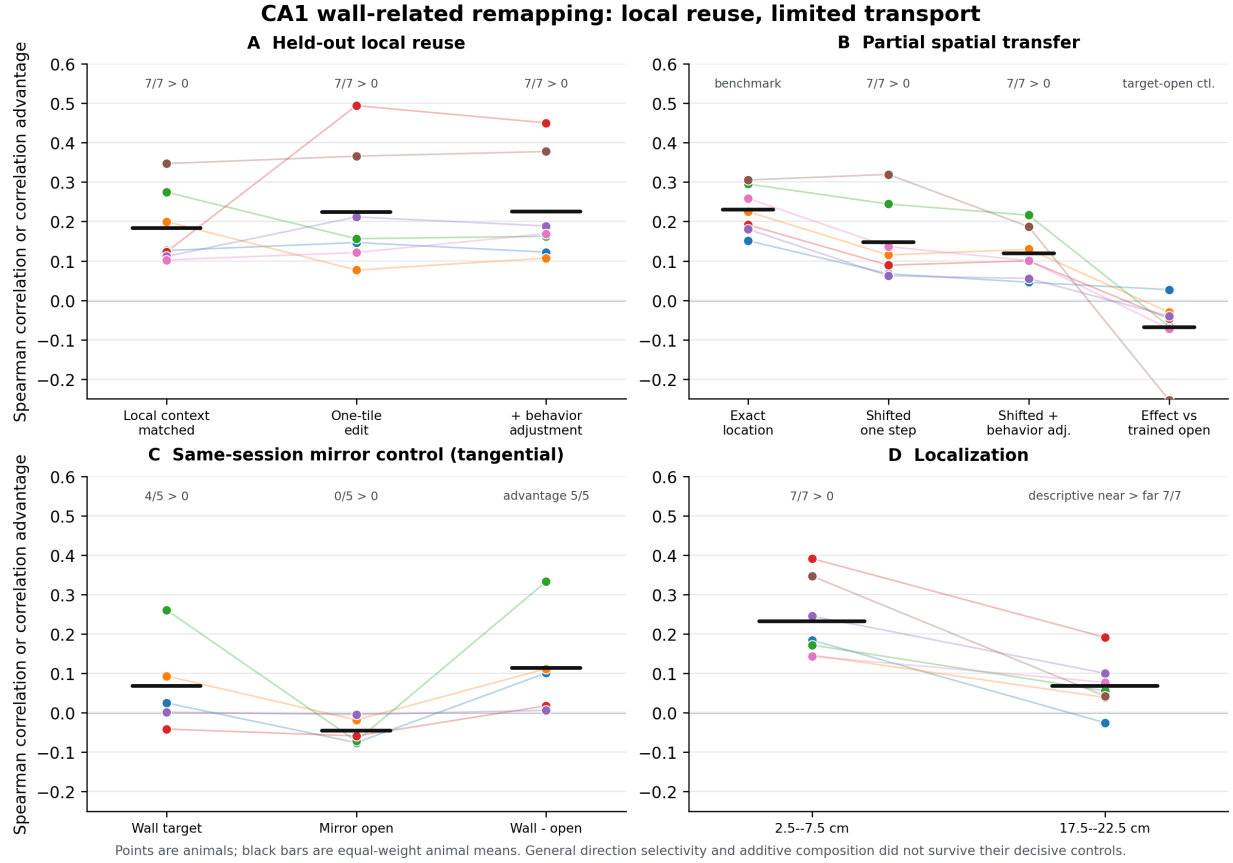


Figure 2: Exact reuse, shifted transfer, and key controls. **A**, wall-minus-open prediction advantages at the same seam, including the single-tile contrast and its behavior-adjusted estimate. **B**, the query-matched exact-location benchmark, the translated source-effect prediction, its behavior-adjusted estimate, and its specificity against a training-derived target-open profile. **C**, the same-session mirror-open control on its supported paired subset: correlations for the wall and mirror-open targets and their wall-minus-open advantage. Only five animals had eligible paired support. **D**, descriptive near and far bands for the single-tile test; the bands were separately eligible and should not be read as a paired estimate. Points are animal means; bars show the across-animal mean.

geometry permits this match only for tangential translations. Although 104 labeled configurations were feasible, paired-support gates retained 72 triplets in five animals; two animals had no admissible control.

On this restricted subset, the globally demeaned source vector favored the wall target over the mirror-open target by 0.114, positive in all five eligible animals (Fig. 2C). Absolute source-to-wall correlation was positive in four of five animals ($\bar{\rho} = 0.068$), and a direct within-session wall-minus-open difference-vector score was positive in four of five ($\bar{\rho} = 0.077$). The raw-rate primary was positive in only three of five. The strongest justified reading is therefore narrow: for the covered demeaned tangential cases, purely symmetric nearby continuity is disfavored. The control does not establish the full seven-animal effect, normal-axis transfer, or raw-rate specificity. The design chronology is process-declared in version control, not an independently timestamped preregistration.

2.6 Identity, direction, and composition controls define the claim’s ceiling

We tested whether cell identity contributed to the shifted prediction by coherently permuting source-cell labels 999 times within each animal while preserving the geometric queries. Descriptive upper-tail fractions were 0.007, 0.001, 0.001, 0.069, 0.005, 0.002, and 0.060. Five animals therefore showed strong identity specificity under this randomization, while two did not. Because the available registered cells impose limited exchangeability, these fractions are diagnostics rather than a population p -value.

Direction controls did not reveal a general, layout-matched normal-direction rule. Same- versus opposite-signed normals for tangential translation favored the same direction in three of six eligible animals. A normal-axis comparison against opposite-facing or overlapping alternatives was positive in four of seven. A less controlled same- versus opposite-away comparison was positive in all seven, but it also changed distance and layout. The reliable statement is therefore transfer between selected geometrically corresponding seams, not rotation- or direction-invariant transport.

A portable representational primitive might also combine additively when two walls are present. The design matrix had rank 10 for 25 possible lattice states, and only two complete two-wall factorials were available; both involved the same bit and donut layouts. In five eligible animals, the demeaned additive predictor reached mean correlation 0.153, below an oracle best-single-component predictor at 0.178. The additive advantage was -0.024 and positive in one of five animals; normalized prediction error was worse in all five. These data provide no positive evidence for linear two-wall composition, but the sparse and repetitive factorial support cannot establish general non-compositionality.

2.7 No support for learning, innateness, or a persistent scar

A transfer-by-experience difference-in-differences estimate was positive under one stored-map specification (mean 0.045, four of four eligible animals). Raw-rate and stricter sensitivity analyses were unstable, including one negative result. Environment order, exposure number, and geometry were not independently varied. We therefore do not infer that experience stabilized the transferred update. For the same reason, the data say nothing about whether the organization was innate rather than learned before imaging.

Finally, we evaluated a prospectively defined former-barrier endpoint in an independent neutral-barrier dataset (Blair et al., 2023b). The primary Pearson-active former-barrier center-minus-control contrast was negative in all leave-one-animal-out summaries (animal mean -0.183 ; one of six individual animals positive). A Spearman sensitivity was near zero (mean 0.001; three of six positive). Thus we found no evidence that a removed neutral barrier left a persistent local representational scar under that endpoint.

3 Discussion

The central result is a graded form of reusability in a hippocampal spatial code. A cell-resolved wall-minus-open response learned from one group of geometries predicted a later response at the same seam, and a separately learned response at a different seam also predicted a later target-wall response. The cross-location prediction was positive in every animal but weaker than the matched exact-location prediction. This combination is the scientific contribution: local boundary-related remapping is neither wholly coordinate-bound nor freely portable.

The finding goes beyond the established observation that boundaries influence place fields. Boundary-vector models explain how fields can arise from inputs tuned to distance and direction

Table 1: Evidence ladder and claim boundary. Values are means across animal-level estimates. Sign counts are descriptive consistency summaries.

Test	Mean	Sign	Supported interpretation
Context-matched exact seam	0.184	7/7	Local wall-open population contrast recurs across layouts.
Single-tile exact seam	0.225	7/7	Exact-location reuse survives a cleaner local counterfactual.
Behavior-adjusted single tile	0.225	7/7	Measured speed, heading, time, and sampling do not explain reuse.
Shifted source effect	0.148	7/7	A cell-resolved wall update contains cross-location information.
Query-matched exact effect	0.230	7/7	Exact reuse exceeds shifted transfer.
Behavior-adjusted shifted effect	0.119	7/7	Measured behavior does not eliminate transfer.
Mirror-open advantage	0.114	5/5	Demeaned tangential subset is not explained by symmetric nearby continuity.
Additive two-wall minus best single	-0.024	1/5	No positive evidence for linear composition in sparse factorial support.
Former-barrier scar	-0.183	1/6	No persistent scar under the independent locked endpoint.

from boundaries (Hartley et al., 2000; Barry et al., 2006), and barrier manipulations have shown local field formation, duplication, and geometry-dependent remapping (Rivard et al., 2004; Lee et al., 2025). Those results make some recurrence plausible. They do not by themselves demonstrate that a signed vector of responses, indexed by the identities of longitudinally registered CA1 cells, generalizes from one physical seam to another under held-out evaluation.

The attenuation relative to exact reuse is equally informative. It argues against describing the result as a universal “wall neuron” vector or a coordinate-free operator. A useful conceptual model is a shared update gated by spatial context:

$$\Delta \mathbf{r}(s, c) = \alpha(s, c) \mathbf{u}_{\text{wall}} + \mathbf{v}(s, c),$$

where $\mathbf{u}_{\text{wall}}$ is a reusable population component, α is a location- and context-dependent gain, and $\mathbf{v}$ contains seam-specific changes. The present analyses establish neither the uniqueness nor the biological mechanism of this decomposition; the equation summarizes the empirical pattern that transfer is nonzero but incomplete.

Several alternative explanations remain. First, smooth or repeated place fields can correlate at nearby seams without an explicit neural operation. The non-overlapping strips, square residuals, trained-open specificity, and mirror-open result constrain this account, but the mirror control is limited to five animals and demeaned tangential translations. Second, unmeasured behavior such as attention, wall inspection, acceleration, or route choice could contribute despite adjustment for measured kinematics. Third, the fixed environment sequence leaves environment-specific order confounded with the single-tile contrast. Fourth, registration and activity thresholds select cells stable enough to be observed across sessions. Fifth, the identity permutation is not an ideal exchangeability test for correlated place-cell populations.

The negative and underpowered controls should not be converted into broad claims. The current design does not show linear wall composition, but it has too few independent factorials to rule composition out. It does not show experience-dependent stabilization, because experience and environment order are aliased. It does not test innateness: every animal had extensive prior

experience with spatial boundaries before the recorded comparisons. The independent neutral-barrier analysis also failed to support a persistent former-barrier scar. These constraints make “limited spatial transfer” a more accurate description than a general reusable mental-map primitive.

The decisive next experiment is compact. Use a symmetric lattice arena with four equivalent candidate wall seams. Randomly assign source and target locations, wall orientation, insertion or removal, and session order within animal. Measure a source wall-open update, then test exact reuse, translated reuse, rotated reuse, and a distance-matched open target in the same later session. Repeat each assignment across early and late exposure blocks, and include a balanced two-wall factorial. This design would independently identify spatial transfer, directional transport, composition, and learning. The present effect sizes provide realistic targets: an exact-location correlation near 0.23, shifted correlation near 0.12–0.15, and a shifted-minus-exact penalty near 0.08.

4 Materials and methods

4.1 Dataset and analysis cohort

We reanalysed the public longitudinal dorsal-CA1 calcium-imaging dataset of [Lee et al. \(2025, 2024\)](#). The tracked analysis cohort contains seven mice, 5,413 neurons, 207 sessions, a familiar outer square, and nine internal-wall geometries arranged on a common lattice. Cell identities were registered across sessions in the source dataset. The original study describes surgery, imaging, behavior, preprocessing, registration, and animal procedures. No new animal experiments were performed.

4.2 Spatial representation and oriented seams

The arena was expressed in a common allocentric coordinate system. A candidate internal-wall location was represented as an oriented seam $s = (p \rightarrow q)$ between adjacent lattice tiles, with a signed normal from tile p to tile q . The binary state $z_{e,s}$ indicated whether environment e blocked that adjacency. Local strips were partitioned into 5 cm spatial bins at defined distances from the seam. Analyses used only bins and registered cells satisfying the relevant occupancy, activity, and common-support gates. Source and target strips in the cross-location analysis shared no 5 cm bin.

4.3 Local response vectors and square residuals

Let $r_{i,e,s}$ denote the local response estimate for registered cell i in environment e at seam s , averaged over eligible spatial bins after the analysis-specific preprocessing. The corresponding bracketing-square estimate was $r_{i,\square(e),s}$. The primary environment-specific response was

$$\delta_{i,e,s} = r_{i,e,s} - r_{i,\square(e),s}.$$

For globally demeaned variants, each vector’s common-cell mean was removed before correlation. This suppresses population-wide rate shifts and makes the score depend on relative cell identities.

4.4 Training profiles and held-out scoring

For a target query (e^*, s^*) , training environments were selected without using the neural response of e^* . At a source seam s , the wall and open profiles were

$$\mu_{W,s} = \text{mean}_{e \in \mathcal{T}: z_{e,s}=1} \delta_{e,s}, \quad \mu_{O,s} = \text{mean}_{e \in \mathcal{T}: z_{e,s}=0} \delta_{e,s},$$

computed on the common registered-cell support. The learned source effect was $\mathbf{d}_s = \mu_{W,s} - \mu_{O,s}$. The held-out outcome was δ_{e^*,s^*} . Scores were Spearman correlations across registered cells. Wall-open advantages compare correlations within the same query and cell support.

The exact-location estimator set $s = s^*$. Its context-matched version restricted training comparisons so the broader internal-wall context was balanced between the wall and open profiles. The reported exact-location benchmark for shifted transfer was recomputed on the exact queries, sessions, cells, and local supports eligible for the shifted analysis.

4.5 Single-tile counterfactual

The u and o geometries differed at one lattice tile: u blocked tiles $\{4, 5\}$, whereas o blocked only tile $\{4\}$. We formed the cell-wise $u - o$ contrast at the focal seam and evaluated wall and open outcomes at that seam in held-out i , l , and t geometries. Near (2.5–7.5 cm) and far (17.5–22.5 cm) bands were screened separately, so their difference is descriptive. For the order-placebo analysis, we formed later-minus-earlier contrasts at unchanged seams with the same processing and compared their signs with the actual $u - o$ contrast.

4.6 Cross-location transfer

For each eligible target seam, the source seam was a non-overlapping lattice seam 25 cm away with the same signed wall normal and admissible wall/open training support. Source training and target outcome sessions were disjoint. The primary shifted score was

$$\rho_{\text{shift}} = \rho_S(\mathbf{d}_{s_{\text{source}}}, \delta_{e^*, s_{\text{target}}}).$$

Secondary specificity scores compared the source effect with a training-derived open profile at the target and compared source-wall with source-open predictors. The exact-location benchmark used $\mathbf{d}_{s_{\text{target}}}$ on the identical held-out query support. Stored-map and behavior-adjusted audits agreed on 405 target queries and 658 source pairs, including their session and seam identities and their counts of common cells and bins. That audit did not hash the complete cell-ID or bin-coordinate arrays.

4.7 Behavior-adjusted response estimates

We estimated activity jointly with spatial-bin indicators and nuisance covariates for instantaneous speed, sine and cosine of heading, second heading harmonics, and elapsed time. Spatial effects were evaluated at a common reference covariate distribution calibrated from the first familiar-square session. The no-time sensitivity omitted elapsed time but retained the other covariates. Target-session activity was used to estimate nuisance coefficients for that target map, but target neural outcomes never entered source-profile training.

4.8 Same-session mirror-open control

Before inspecting the neural endpoint within the project workflow, we defined a mirror-open seam in the same target session as each target wall. It matched the source-wall pair in translation distance, translation axis, signed normal, relative spatial bins, and registered cells. Both source-to-target nearest-bin distances were 5 cm. Only tangential translations admitted this exact geometric pairing. Primary scoring used globally demeaned source and target vectors. Eligibility required a complete source, wall-target, and open-target triplet on common support.

4.9 Cell-identity permutation

For each animal, we generated 999 coherent random mappings of source-cell identities while preserving query geometry, target vectors, support sizes, and the mapping across all queries in that permutation. The descriptive upper-tail fraction was $(1 + n_{\text{null} \geq \text{observed}})/(1000)$. Correlations and registration constraints limit exchangeability; the fractions were used as identity-specificity diagnostics rather than inferential population p -values.

4.10 Direction, composition, and experience analyses

Direction controls compared eligible source-target pairs with same or opposite signed wall normals while documenting translation axis, distance, overlap, and layout matching. No single comparison simultaneously matched all of these features across the full cohort.

For composition, we identified complete wall-state factorials in which two single-wall effects and their joint state could be estimated. The additive prediction was the sum of the two single effects. It was compared with an oracle best-single-component predictor using correlation and normalized prediction error. Only two repetitive factorials and five eligible animals were available.

Experience analyses contrasted transfer changes across exposure blocks using difference-in-differences variants. Because geometry, order, and exposure were not independently randomized, these estimates were treated as exploratory sensitivity analyses and not as evidence for learning.

4.11 Independent former-barrier analysis

We used the public neutral-barrier data from Blair et al. (2023b) to test whether the former barrier center retained an excess neural similarity relative to matched control locations after removal. The primary endpoint was a Pearson correlation among active cells, averaged as a former-barrier center-minus-control contrast. Its definition was written before the neural endpoint was run in the local project workflow. Spearman scoring and leave-one-animal-out summaries were sensitivities.

4.12 Statistical reporting and reproducibility

Animals were treated as the biological units. We report animal-level means, sign counts, query counts, and explicitly labeled descriptive permutation fractions. No cell-level or query-level observation was presented as an independent animal replicate. The analyses are exploratory: neither the original experiment nor the present reanalysis was preregistered for the cross-location hypothesis.

All figure values derive from tracked machine-readable source files. The repository contains scripts, eligibility ledgers, result summaries, design locks, and an inspectable Git history. The principal manuscript figure is generated from its tracked source-data CSV.

Data and code availability

The CA1 source data are available through the archive cited by Lee et al. (2024). The independent miniscope data are available through the archive cited by Blair et al. (2023a). Analysis code, derived source tables, audit reports, and manuscript sources are publicly available at <https://github.com/asim-v/ca1-wall-update-transfer>. The public release excludes the original raw recordings, which remain available from their source archives.

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Author contributions

To be completed using the CRediT taxonomy.

Competing interests

The authors declare no competing interests, subject to author confirmation.

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